

Evaluating and Equivalent Expressions

MA.8.NSO.1.3

I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

So far: every law, every kind of base, every kind of exponent.

Today: use all of it at once — and rewrite bases to compare.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
2. Find the value: $\frac{1}{5^{-2}}$
3. Find the value: $\frac{10^6}{10^4}$
4. Write 64 as a power of 2, of 4, and of 8.

Warm-Up

Answers.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
 $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$
2. Find the value: $\frac{1}{5^{-2}}$ **25**
3. Find the value: $\frac{10^6}{10^4}$ **100**
4. Write 64 as a power of 2, of 4, and of 8. **$2^6, 4^3, 8^2$**

Notes

I. Two students, two plans, one value

$$\text{Melanie: } \frac{5^{-2} \cdot 5^5}{5} = \frac{5^3}{5} = 5^2 = 25$$

$$\text{Adonis: } \frac{5^{-2} \cdot 5^5}{5} = \frac{\frac{1}{25} \cdot 3125}{5} = \frac{125}{5} = 25$$

Apply the laws first, evaluate last. The value is the same; the arithmetic is smaller.

Notes

Copy the worked line.

II. Every law, one expression

$$\frac{(6^2)^{-3}}{6^{-5}} = \frac{6^{-6}}{6^{-5}} = 6^{-6 - (-5)} = 6^{-1} = \frac{1}{6}$$

Find the step that broke:

$$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} = \left(\frac{1}{5}\right)^{-8} \quad \text{— wrong: multiplied; should be } \left(\frac{1}{5}\right)^2$$

$$\left(\frac{3}{2}\right)^{-3} = \frac{2^{-3}}{3^{-3}} \quad \text{— wrong: flipped, kept the minus; should be } \left(\frac{2}{3}\right)^3$$

Notes

III. Different bases, same value — rewrite to a common base

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

$$8^5 = (2^3)^5 = 2^{15} \qquad 9^4 = (3^2)^4 = 3^8$$

$$2^3 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$$

Notes

IV. Testing equivalence: Xavier's claim

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3} = 2^{-4} \cdot 2^{-3} = 2^{-7} \qquad \left(\frac{1}{2}\right)^4 \cdot 4^{-1} = 2^{-4} \cdot 2^{-2} = 2^{-6}$$

Same base, then compare exponents.

$2^{-7} \neq 2^{-6}$ — so Xavier is wrong.

Example 1

Evaluate.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3$$

Find the value.

Example 1

Worked.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3 = \left(\frac{4}{3}\right)^2 \cdot 3 \quad \text{negative exponent: flip}$$

$$\frac{16}{9} \cdot 3 = \frac{16}{3} \quad \text{power of a quotient, then multiply}$$

Answer: 16/3

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

Your Turn

Answer.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

$$(5/2)^2 \cdot 2 = 25/4 \cdot 2$$

$$\textbf{Answer: } \frac{25}{2}$$

Example 2

Equivalent or not?

Are $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ equivalent?

Rewrite both with the bases 3 and 2, then compare.

Example 2

Worked.

$$3^2 \cdot 6^3 = 3^2 \cdot 3^3 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{power of a product, product of powers}$$

$$\frac{(3 \cdot 2)^5}{2^2} = \frac{3^5 \cdot 2^5}{2^2} = 3^5 \cdot 2^3 \quad \text{power of a product, quotient of powers}$$

$$3^5 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{same bases, same exponents}$$

Answer: Yes — both are $3^5 \cdot 2^3 = 1944$

Your Turn

Same steps, your numbers. Boards up when done.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

Your Turn

Answer.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

both are $2^5 \cdot 5^3$

Answer: Yes — $2^5 \cdot 5^3 = 4000$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

Answer it.

Find the value.

Whiteboards • Question 1 of 9

Answer.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

$(-7)^5$ over $(-7)^2$, so $(-7)^3$

Answer: -343

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

$$-3 - (-2) = -1, \text{ so } (3/4)^{-1}$$

$$\textbf{Answer: } \frac{4}{3}$$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

$$2^{-3} \cdot 2^4 \text{ over } 2^3 = 2^{-2}$$

$$\text{Answer: } \frac{1}{4}$$

Whiteboards • Question 4 of 9

Boards up on three.

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

Answer it.

Whiteboards • Question 4 of 9

Answer.

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

$$8 = 2^3, \text{ so } 8^5 = (2^3)^5$$

Answer: The same: $8^5 = (2^3)^5 = 2^{15}$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^4$$

Which expression is equivalent?

A. 3^{16}

B. 3^4

C. 3^8

D. 3^6

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^4$$

Which expression is equivalent?

A. 3^{16}

B. 3^4

C. 3^8

D. 3^6

Answer: C — 3^8

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^7

C. 2^{-5}

D. 2^{-12}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^7

C. 2^{-5}

D. 2^{-12}

Answer: A — 2^{-7}

Whiteboards • Question 7 of 9

Boards up on three.

$$2^3 \cdot 4^3$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 7 of 9

Answer.

$$2^3 \cdot 4^3$$

$$4^3 = (2^2)^3 = 2^6$$

Answer: 2^9

Whiteboards • Question 8 of 9

Boards up on three.

$$4^7 \cdot 4^{-2}$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 8 of 9

Answer.

$$4^7 \cdot 4^{-2}$$

$$4^5 = (2^2)^5$$

Answer: 2^{10}

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree?

Write your answer in sentences.

Rewrite each with base 2, and say what you find.

Whiteboards • Question 9 of 9

Answer.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent

because both can be written as 2^{-7} . Do you agree?

2^{-7} and 2^{-6}

Answer: No — the second is 2^{-6} , not 2^{-7}

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — 6T7

Identify equivalent expressions involving exponents II — QDM